Mathematical and Computational Methods

Lecture 1: Foundations: numbers, sets and functions

This opening lecture assembles the vocabulary and objects on which the whole module rests: the number systems we compute in, the set, interval and summation notation we write with, and—above all—the idea of a *function* together with the catalogue of elementary functions used throughout the course. The aim is fluency with the language, not difficulty: almost everything here will be used in every later lecture.

Where this fits. Nothing is assumed from a previous lecture; this is the starting point. The chain of number systems ends at the complex numbers $\mathbb{C}$, which we build in Lectures 2–3, and the exponential, trigonometric and hyperbolic functions introduced here reappear there in the polar and Euler forms. The notions of domain, monotonicity and symmetry set up the calculus of Lectures 7–10, and the ReLU and sigmoid functions return in the optimisation and data-science material.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Use set, interval and Σ/Π notation, and solve inequalities.
- Find a function's maximal domain and range, and compose and decompose functions.
- Classify functions (injective/surjective/bijective, even/odd/periodic) and find inverses.
- Sketch and transform the elementary functions, including $|x|$, sgn, H and ReLU.
- State $\cosh^2 x - \sinh^2 x = 1$ and read the sigmoid $\sigma(x)$ as a probability.

1 Numbers, sets and notation

1.1 Number systems

Mathematics repeatedly enlarges its number system whenever an operation leads outside the numbers currently available. We begin with the *natural numbers* used for counting and successively repair each shortcoming.

Definition 1.1 (The standard number systems).

- $\mathbb{N} = \{1, 2, 3, \dots\}$ — the *natural numbers* (counting). We write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ when 0 is included.
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ — the *integers*; closed under subtraction.
- $\mathbb{Q} = \left\{ \dfrac{p}{q} \;:\; p \in \mathbb{Z},\ q \in \mathbb{N} \right\}$ — the *rationals*; closed under division by nonzero rationals.
- $\mathbb{R}$ — the *real numbers*; informally the "number line with no gaps" (this completeness is made precise in Analysis). It contains $\mathbb{Q}$ together with numbers such as $\sqrt{2}$, π and e that are not fractions.
- $\mathbb{C} = \{a + b\,\mathrm{i} \;:\; a, b \in \mathbb{R}\}$, where i is a symbol with $\mathrm{i}^2 = -1$ — the *complex numbers*, the subject of Lectures 2–3.

These sit inside one another, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Remark. Each extension removes an obstruction. Subtraction can leave $\mathbb{N}$ (e.g. $3 - 5$), so we pass to $\mathbb{Z}$; division can leave $\mathbb{Z}$ (e.g. $1 \div 2$), so we pass to $\mathbb{Q}$; taking limits or roots can leave $\mathbb{Q}$ (e.g. $\sqrt{2}$), so we pass to $\mathbb{R}$; and the equation $x^2 + 1 = 0$ has no real solution, which forces the step to $\mathbb{C}$.

1.2 Rational and irrational numbers

A real number is *rational* if it can be written as a fraction of integers and *irrational* otherwise. A second, independent classification cuts across this one.

Definition 1.2 (Algebraic and transcendental numbers). A real (or complex) number is *al-*

gebraic if it is a root of some nonzero polynomial with integer coefficients, and *transcendental* otherwise.

Every number is algebraic or transcendental. Every rational is algebraic (p/q is a root of $qx - p$), and so are some irrationals: $\sqrt{2}$ is a root of $x^2 - 2$, and $\sqrt[3]{5}$ a root of $x^3 - 5$. The transcendentals, such as π and e, are the numbers that are roots of *no* integer polynomial; they are necessarily irrational, though they can still be approximated by rationals such as $\pi \approx \frac{22}{7}$. The next example shows the standard way to prove a number irrational.

Example 1.1 ($\sqrt{2}$ *is irrational*). Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{N}$ sharing no common factor. Squaring gives $p^2 = 2q^2$, so p^2 is even, hence p is even, say $p = 2k$. Then $4k^2 = 2q^2$, i.e. $q^2 = 2k^2$, so q^2 is even and q is even too. But then p and q share the factor 2, contradicting our assumption. Therefore no such fraction exists and $\sqrt{2}$ is irrational.

1.3 Intervals and infinity

For $a < b$ in $\mathbb{R}$ we use the standard interval notation

$$(a, b) = \{\, x \in \mathbb{R} \,:\, a < x < b \,\}, \qquad [a, b] = \{\, x \in \mathbb{R} \,:\, a \le x \le b \,\},$$

with the half-open intervals $[a, b)$ and $(a, b]$ defined analogously: a square bracket includes the endpoint, a round bracket excludes it. Unbounded intervals use the symbols $\pm\infty$, for instance

$$[a, \infty) = \{\, x \in \mathbb{R} \,:\, x \ge a \,\}, \qquad (-\infty, b) = \{\, x \in \mathbb{R} \,:\, x < b \,\}.$$

Note. The symbol ∞ is *not* a real number; it is shorthand for "grows without bound". An interval boundary at ∞ is therefore always written with a round bracket, never a square one.

1.4 Sets and set-builder notation

A *set* is a collection of distinct objects. Small sets are written by listing their elements, as in $\{1, 2, 3\}$; larger or infinite sets are described by a defining property using *set-builder notation*,

$$\{\, x \,:\, P(x) \,\} \qquad \text{"the set of all } x \text{ such that } P(x) \text{ holds".}$$

For example $\{\, x \in \mathbb{R} \,:\, x^2 < 4 \,\} = (-2, 2)$. The basic operations are union $A \cup B$ (elements in A *or* B), intersection $A \cap B$ (in A *and* B), difference $A \setminus B$ (in A but not B), and the empty set $\varnothing$ containing no elements.

1.5 Sum and product notation

Long sums and products are abbreviated with the Greek capitals Σ (sigma) and Π (pi):

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n, \qquad \prod_{k=1}^{n} a_k = a_1 \cdot a_2 \cdots a_n.$$

Here k is a *dummy index* running over the stated range. Two facts worth remembering are the arithmetic-series sum and the factorial,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \prod_{k=1}^{n} k = n!\,.$$

Example 1.2 *(Evaluating a sum)*. The sum of the first five odd numbers is

$$\sum_{k=1}^{5}(2k-1) = 1+3+5+7+9 = 25.$$

Splitting the sum confirms this in general:

$$\sum_{k=1}^{n}(2k-1) = 2\sum_{k=1}^{n}k - \sum_{k=1}^{n}1 = 2\cdot\frac{n(n+1)}{2} - n = n^2,$$

and indeed $5^2 = 25$.

1.6 Inequalities

Inequalities obey rules close to those for equations, with one crucial exception.

Note (Rules for inequalities). For real numbers one may add or subtract the same quantity on both sides freely, and multiply or divide both sides by a *positive* number. Multiplying or dividing by a *negative* number **reverses** the inequality sign. Also, for $a > 0$,

$$|x| \le a \iff -a \le x \le a, \qquad |x| \ge a \iff x \le -a \text{ or } x \ge a.$$

Example 1.3 *(A quadratic and an absolute-value inequality)*. To solve $x^2 - x - 6 < 0$, factor: $(x-3)(x+2) < 0$. The product is negative exactly between the roots, so the solution is $-2 < x < 3$, i.e. the interval $(-2, 3)$.

To solve $|2x-1| \le 3$, remove the modulus: $-3 \le 2x - 1 \le 3$, hence $-2 \le 2x \le 4$ and finally $-1 \le x \le 2$, i.e. $[-1, 2]$.

2 Functions and their properties

2.1 What a function is

Definition 2.1 (Function, domain, codomain, range). A *function* $f\colon A \to B$ is a rule assigning to *each* element $x \in A$ exactly one element $f(x) \in B$. The set A is the *domain*, B is the *codomain*, and the *range* (or image) is the set of values actually attained,

$$f(A) = \{\, f(x) \;:\; x \in A \,\} \subseteq B.$$

When a function is given only by a formula, its *maximal* (or natural) domain is the largest set of real inputs for which the formula makes sense—excluding, for instance, division by zero or even roots of negative numbers.

Example 2.1 *(Maximal domain)*. For $f(x) = \dfrac{\sqrt{x+3}}{x-1}$ we need $x + 3 \ge 0$ (for the square root) and $x - 1 \neq 0$ (no division by zero). The maximal domain is therefore

$$\{\, x \in \mathbb{R} \;:\; x \geq -3,\ x \neq 1 \,\} = [-3, 1) \cup (1, \infty).$$

Example 2.2 (*Finding a range*). For $g(x) = 2 - \sqrt{4 - x}$ the maximal domain is $x \leq 4$. As x runs over $(-\infty, 4]$ the inner term $\sqrt{4 - x}$ takes every value in $[0, \infty)$, so $-\sqrt{4 - x}$ ranges over $(-\infty, 0]$ and $g(x) = 2 - \sqrt{4 - x}$ ranges over $(-\infty, 2]$. The maximum value 2 is attained at $x = 4$, so the range is $(-\infty, 2]$.

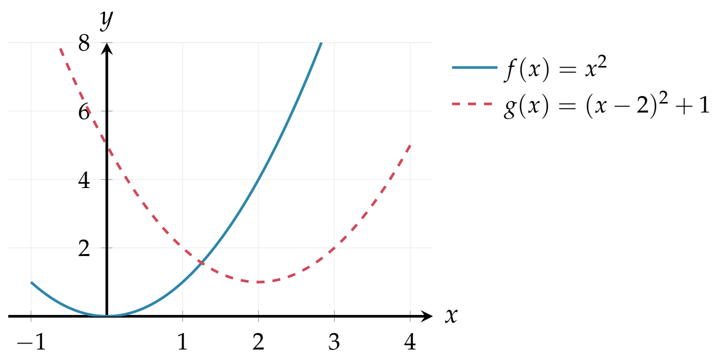
Note. A formula defines a function only if it gives a *single* output for each input. Graphically this is the *vertical line test*: every vertical line meets the graph at most once.

2.2 Graphs and graph transformations

The *graph* of f is the set of points $\{\, (x, f(x)) \;:\; x \in A \,\}$ in the plane. Once the graph of a function f is known, the graphs of simple algebraic modifications follow by shifting, stretching and reflecting.

Transformation	Formula	Effect on the graph
Vertical shift	$f(x) + c$	up by c (down if $c < 0$)
Horizontal shift	$f(x - c)$	right by c (left if $c < 0$)
Vertical stretch	$a\,f(x)$	scale by factor a from the x-axis (for $a > 0$; if $a < 0$ it also reflects in the x-axis)
Horizontal scaling	$f(bx)$	by factor $1/b$: compress to the y-axis if $b > 1$, stretch if $0 < b < 1$
Reflection in x-axis	$-f(x)$	flip top-to-bottom
Reflection in y-axis	$f(-x)$	flip left-to-right

For instance the graph of $g(x) = (x - 2)^2 + 1$ is the parabola $y = x^2$ shifted two units right and one unit up:



2.3 Composition of functions

Definition 2.2 (Composition). Given f and g, the *composite* $f \circ g$ is defined by $(f \circ g)(x) = f(g(x))$: first apply g, then f. Its domain consists of those x in the domain of g for which $g(x)$ lies in the domain of f.

Composition is generally *not* commutative: $f \circ g$ and $g \circ f$ usually differ.

Example 2.3 *(Composition, and decomposition)*. Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$. Then

$$(f \circ g)(x) = \sqrt{1 - x^2} \quad (\text{domain } [-1, 1]), \qquad (g \circ f)(x) = 1 - x \quad (\text{domain } x \geq 0),$$

so the order matters, and so does the domain. Conversely, recognising a function as a composite is a key skill: $h(x) = (2x+1)^5$ is $f \circ g$ with $g(x) = 2x + 1$ and $f(u) = u^5$. We will use exactly this decomposition for the chain rule in Lecture 7.

2.4 Injective, surjective and bijective functions

Definition 2.3 (Three properties of a function $f \colon A \to B$). f is *injective* (one-to-one) if different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. It is *surjective* (onto) if its range is all of B. It is *bijective* if it is both injective and surjective.

Note. Injectivity is the *horizontal line test*: every horizontal line meets the graph at most once.

Example 2.4 *(The same rule, different verdicts)*. As a map $f \colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is *not* injective ($f(-1) = f(1) = 1$) and *not* surjective (it never takes negative values). If instead we restrict the domain and codomain to $f \colon [0, \infty) \to [0, \infty)$, the same rule becomes bijective. This illustrates that injectivity, surjectivity and bijectivity are properties of the rule *together with* its stated domain and codomain.

2.5 Inverse functions and monotonicity

Definition 2.4 (Inverse function). If $f \colon A \to B$ is bijective, its *inverse* $f^{-1} \colon B \to A$ is the function that undoes it: $f^{-1}(y) = x$ exactly when $f(x) = y$. Equivalently $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

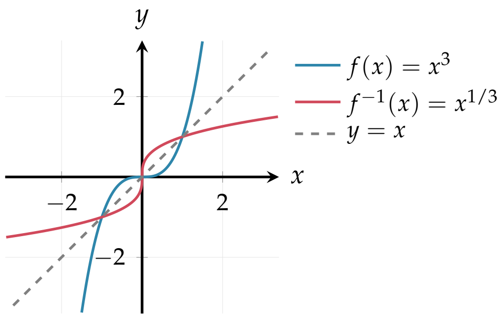
A function is *strictly increasing* if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$, and *strictly decreasing* if the inequality reverses; either way it is *strictly monotonic*. A strictly monotonic function is automatically injective; restricting its codomain to its range then makes it bijective, so it has an inverse on that range—the most common way to guarantee one exists. To find an inverse algebraically, write $y = f(x)$ and solve for x in terms of y.

Example 2.5 *(Finding an inverse)*. Let $f(x) = \sqrt{x - 1}$ on the domain $[1, \infty)$, with range $[0, \infty)$. Setting $y = \sqrt{x - 1}$ and solving gives $x = y^2 + 1$, so

$$f^{-1}(x) = x^2 + 1, \qquad \text{domain } [0, \infty), \ \text{range } [1, \infty).$$

Note how domain and range swap between f and f^{-1}.

Note. The graph of f^{-1} is the mirror image of the graph of f in the line $y = x$, because reflecting in $y = x$ swaps the roles of input and output. The figure below shows this for $f(x) = x^3$, which is bijective on all of $\mathbb{R}$: solving $y = x^3$ gives $x = y^{1/3}$, so $f^{-1}(x) = x^{1/3}$ for every real x (no domain restriction needed).



2.6 Symmetry: even, odd and periodic functions

Definition 2.5 (Symmetry). A function with a domain symmetric about 0 is *even* if $f(-x) = f(x)$ for all x (graph symmetric about the y-axis) and *odd* if $f(-x) = -f(x)$ for all x (graph symmetric about the origin). A function is *periodic* with period $T > 0$ if $f(x+T) = f(x)$ for all x; for a non-constant function the smallest such T is *the* period.

For example x^2 and $\cos x$ are even, x^3 and $\sin x$ are odd, and $\sin x$ is periodic with period 2π. Most functions are neither even nor odd. Recognising symmetry halves the work of sketching a graph and, later, of computing integrals.

3 Elementary functions and their graphs

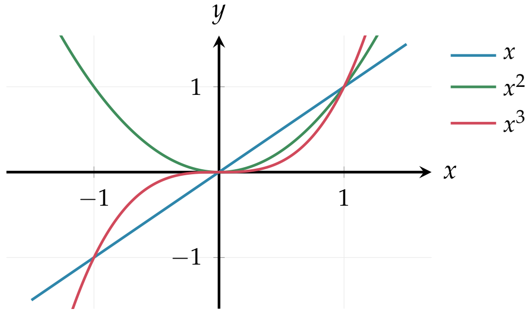
The functions below, with their compositions, inverses and the four arithmetic operations, make up the *elementary functions*—the working vocabulary for the rest of the module.

3.1 Power, polynomial and rational functions

A *power function* is $x \mapsto x^n$. For positive integer n it is even when n is even (symmetric, non-negative) and odd when n is odd. A *polynomial* of degree n is a finite sum

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

with domain all of $\mathbb{R}$; for large $|x|$ its sign and growth are governed by the leading term $a_n x^n$. A *rational function* is a quotient $p(x)/q(x)$ of two polynomials; its maximal domain excludes the zeros of the denominator q, near which the graph typically has vertical asymptotes.



Example 3.1 ***(Domain of a rational function).*** For

$$r(x) = \frac{x+1}{x^2-4} = \frac{x+1}{(x-2)(x+2)}$$

the denominator vanishes at $x = \pm 2$, so the maximal domain is $\mathbb{R} \setminus \{-2, 2\}$, and the lines $x = -2$ and $x = 2$ are vertical asymptotes.

3.2 Exponential and logarithmic functions

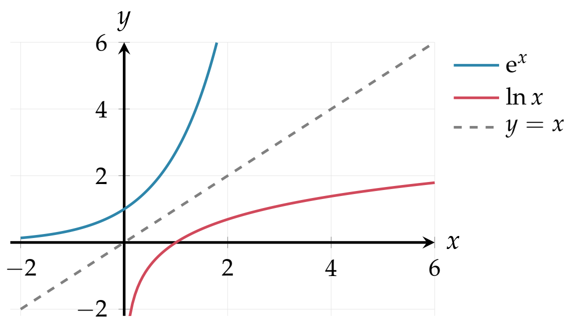
For a base $a > 0$, $a \neq 1$, the *exponential function* a^x has domain $\mathbb{R}$ and range $(0, \infty)$, and obeys the index laws

$$a^{x+y} = a^x a^y, \qquad (a^x)^y = a^{xy}, \qquad a^0 = 1, \qquad a^{-x} = \frac{1}{a^x}.$$

The most important base is Euler's number $\mathrm{e} \approx 2.71828$, giving the *natural exponential* e^x. Its inverse is the *natural logarithm* $\ln = \log_{\mathrm{e}}$, with domain $(0, \infty)$, range $\mathbb{R}$, and the laws

$$\ln(xy) = \ln x + \ln y, \quad \ln\frac{x}{y} = \ln x - \ln y, \quad \ln(x^r) = r\ln x, \quad \log_a x = \frac{\ln x}{\ln a}.$$

Because ln inverts e^x, the two graphs are reflections of one another in the line $y = x$.

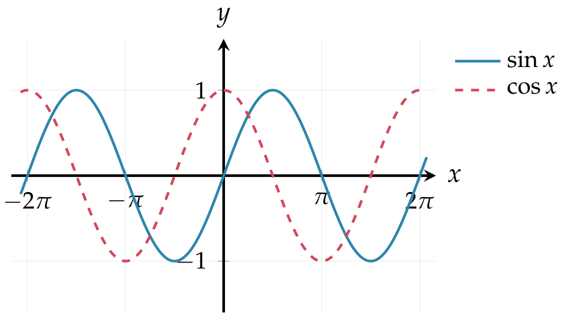


Example 3.2 ***(Solving with exponentials and logarithms).*** To solve $2^x = 10$, take natural logs: $x \ln 2 = \ln 10$, so $x = \dfrac{\ln 10}{\ln 2} = \log_2 10 \approx 3.32$. The log laws also simplify expressions directly, e.g. $\ln 8 - \ln 2 = \ln \frac{8}{2} = \ln 4$.

3.3 Trigonometric and inverse-trigonometric functions

At university angles are measured in *radians*; recall $x^\circ = \frac{\pi}{180}x$ radians, so $180^\circ = \pi$. The functions sin and cos have domain $\mathbb{R}$, range $[-1, 1]$ and period 2π; sin is odd and cos is even. The standard values are worth memorising.

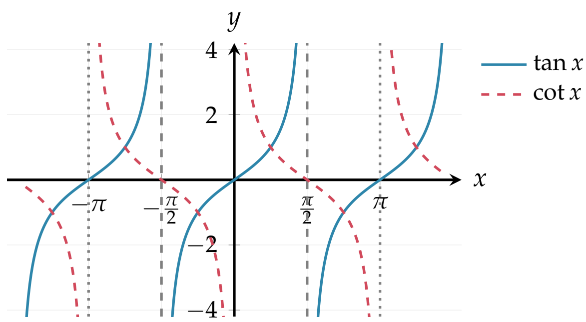
θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0



The remaining trigonometric functions are built from sin and cos. The *tangent* and *cotangent* are

$$\tan x = \frac{\sin x}{\cos x}, \qquad \cot x = \frac{\cos x}{\sin x} = \frac{1}{\tan x},$$

alongside the reciprocals $\sec x = 1/\cos x$ and $\csc x = 1/\sin x$. Both tan and cot are odd, take every real value (range $\mathbb{R}$) and are periodic with period π. Their domains exclude the zeros of the denominator: tan is undefined where $\cos x = 0$, i.e. at $x = \frac{\pi}{2} + k\pi$, and cot is undefined where $\sin x = 0$, i.e. at $x = k\pi$ (with $k \in \mathbb{Z}$); at each excluded point the graph has a vertical asymptote (dashed for tan, dotted for cot below).



We will lean on a handful of identities, recapped here for reference and used heavily in Lecture 3:

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta,$$

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta, \qquad \tan(-\theta) = -\tan\theta,$$

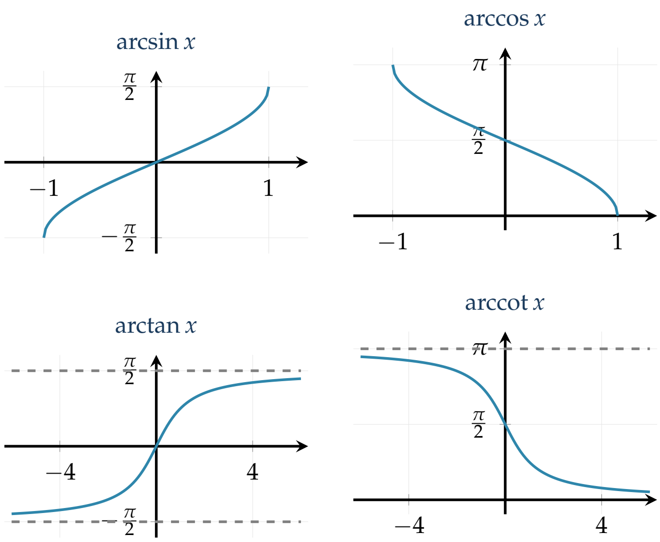
$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta, \qquad \cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$

Inverse trigonometric functions

Being periodic, the trigonometric functions are far from injective, so each is made invertible by restricting it to a single monotonic *principal* branch. This gives the inverse functions

$$\arcsin\colon [-1,1] \to [-\tfrac{\pi}{2}, \tfrac{\pi}{2}], \qquad \arccos\colon [-1,1] \to [0,\pi],$$
$$\arctan\colon \mathbb{R} \to (-\tfrac{\pi}{2}, \tfrac{\pi}{2}), \qquad \operatorname{arccot}\colon \mathbb{R} \to (0,\pi),$$

where arcsin inverts sin on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, arccos inverts cos on $[0,\pi]$, arctan inverts tan on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and arccot inverts cot on $(0,\pi)$. The two tangent-type inverses are bounded: arctan has horizontal asymptotes $y = \pm\frac{\pi}{2}$, and arccot has $y = 0$ and $y = \pi$. As with any inverse, the graph of each is the reflection of its restricted original in the line $y = x$; all four are drawn below.



Example 3.3 ***(A trigonometric equation).*** Solve $\cos x = \frac{1}{2}$ for $x \in [0, 2\pi)$. From the table $\arccos\frac{1}{2} = \frac{\pi}{3}$, and since cosine is even and 2π-periodic the solutions in the range are $x = \frac{\pi}{3}$ and $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$.

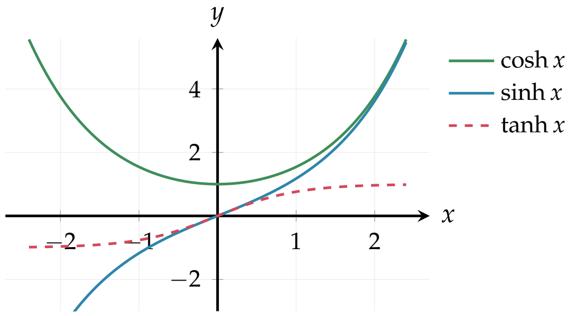
Example 3.4 ***(Tangent and its inverse).*** Solve $\tan x = 1$ for $x \in [0, 2\pi)$. The principal value is $\arctan 1 = \frac{\pi}{4}$, and since tan has period π the solutions in the range are $x = \frac{\pi}{4}$ and $x = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$. Note the distinction: $\arctan 1$ returns only the single principal value in $(-\frac{\pi}{2}, \frac{\pi}{2})$, whereas the equation $\tan x = 1$ has infinitely many solutions.

3.4 Hyperbolic functions (brief)

Built from the exponential, the *hyperbolic* functions are

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}.$$

Here sinh is odd and cosh is even, $\cosh x \geq 1$ for all x, and tanh is odd with horizontal asymptotes $y = \pm 1$. They satisfy the identity below, the hyperbolic analogue of $\sin^2 + \cos^2 = 1$.



Example 3.5 ***(The fundamental identity).*** Directly from the definitions,

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4} = \frac{4\,e^x e^{-x}}{4} = 1.$$

Remark. We keep hyperbolic functions brief here. Their deeper meaning—and the reason cosh and cos look so alike—emerges in Lecture 3, where Euler's formula gives $\cos(ix) = \cosh x$. They reappear once more as standard derivatives in Lecture 7.

3.5 Piecewise-defined functions

A function may be specified by different formulas on different parts of its domain. Several such functions are central to physics and to data science.

Definition 3.1 (Four piecewise functions).

$$\text{absolute value:} \quad |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases} \qquad \text{sign:} \quad \operatorname{sgn}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0, \end{cases}$$

$$\text{Heaviside step:} \quad H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad \text{ReLU:} \quad \operatorname{ReLU}(x) = \max(0, x).$$

A fifth function belongs in this section for a different reason. The *logistic* (or *sigmoid*) function is *not* piecewise—it is a single smooth formula—but it is a close cousin of the step and ReLU and one of the most important functions in data science, so it earns its place here.

Definition 3.2 (The logistic (sigmoid) function).

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

It is smooth and strictly increasing, with $\sigma(0) = \frac{1}{2}$, range $(0, 1)$ and limits $\sigma(x) \to 1$ as $x \to \infty$ and $\sigma(x) \to 0$ as $x \to -\infty$; it is thus a smooth approximation of the Heaviside step H, and it has the symmetry $\sigma(-x) = 1 - \sigma(x)$.

A probability from any number. Because σ sends every real number into the open interval $(0, 1)$, its output can be read as a *probability* —which is what makes it ubiquitous in data science. In *logistic regression*, a real-valued score s assembled from the data (typically a weighted sum of measured features) is turned into an estimated probability $\sigma(s)$ that an item belongs to a class: a large positive s gives a probability close to 1, a large negative s one close to 0, and $s = 0$ the noncommittal $\frac{1}{2}$. The symmetry $\sigma(-x) = 1 - \sigma(x)$ is exactly the statement that the probabilities of the two classes sum to 1.

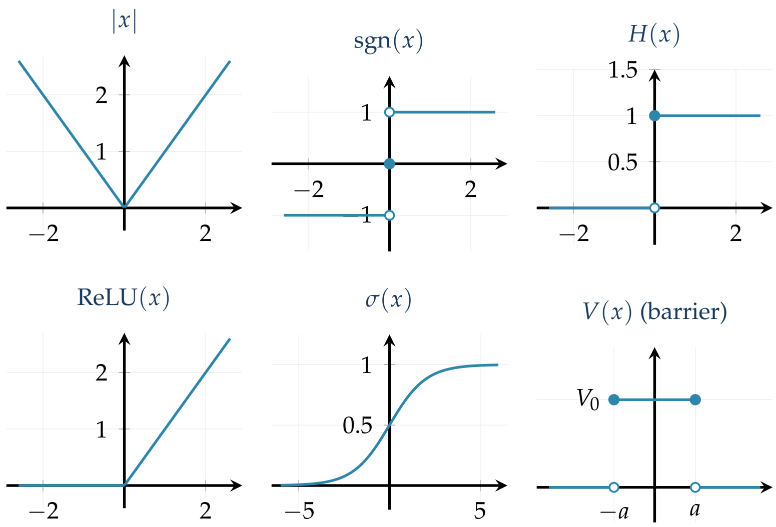
Returning to genuinely piecewise functions, the physics students should meet one more: the *rectangular potential barrier*, a function that is constant on a finite interval and zero outside it.

Definition 3.3 (Rectangular potential barrier). For a height $V_0 > 0$ and width $2a > 0$,

$$V(x) = \begin{cases} V_0, & -a \le x \le a, \\ 0, & \text{otherwise.} \end{cases}$$

This is the finite square barrier of quantum mechanics: a particle of energy below V_0 has a nonzero probability of *tunnelling* through it, which classical mechanics forbids. It is also a basic building block for assembling more complicated potentials out of simple pieces.

The six functions are drawn individually below (taking $V_0 = 1$, $a = 1$ for the barrier). A filled dot marks an included endpoint and an open dot an excluded one, so the jumps in sgn, H and V are unambiguous.



Remark. ReLU and σ are the two best-known *activation functions* in neural networks: the data-science students will meet them again as the nonlinearities between layers, and we will differentiate σ in Lecture 8, where the tidy fact $\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big)$ proves useful.

Example 3.6 *(Reading a piecewise definition).* $\mathrm{ReLU}(x) = \max(0, x)$ can be written piecewise as 0 for $x < 0$ and x for $x \ge 0$; equivalently, in closed form, $\mathrm{ReLU}(x) = \frac{1}{2}\big(x + |x|\big)$.

Summary

Concept	**Key idea**
Number systems	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; each extension repairs an operation that escapes the previous system.
Algebraic vs transcendental	Algebraic = root of an integer polynomial (every rational, and some irrationals like $\sqrt{2}$); transcendental (π, e) otherwise.
Notation	Intervals $[a, b]$, (a, b); set-builder $\{ x : P(x) \}$; sums/products $\sum$, $\prod$; inequality rules (a negative multiplier flips the sign).
Function	A rule $f\colon A \to B$, one output per input; domain, codomain, range, maximal domain.
Transformations	$f(x) + c$, $f(x - c)$, $af(x)$, $f(bx)$, $-f(x)$, $f(-x)$ shift, stretch and reflect the graph.
Composition	$(f \circ g)(x) = f(g(x))$; order matters; decomposition prepares the chain rule.
Inj./surj./bij.	One-to-one / onto / both; bijective $\iff$ invertible; strict monotonicity $\Rightarrow$ injective.
Inverse	Swap-and-solve; domain and range swap; graph reflects in $y = x$.
Symmetry	Even $f(-x) = f(x)$, odd $f(-x) = -f(x)$, periodic $f(x + T) = f(x)$.
Elementary functions	Power/polynomial/rational, e^x and ln, the trigonometric and inverse-trig functions, a brief look at sinh, cosh, tanh, the piecewise $\lvert x \rvert$, sgn, H, ReLU, and the smooth sigmoid σ.

Next lecture: the final step in the number chain—the complex numbers $\mathbb{C}$*—their algebra, the Argand diagram, and the polar form.*

Companion material: a separate *Exercise Sheet 1* (with solutions) develops the practice problems for this unit.